

Kani's lemma from Clifford algebra

Tomoki Moriya

Information Technology R&D Center, Mitsubishi Electric, 5-1-1, Ofuna,
Kamakura, 2470056, Japan.

Contributing authors: Moriya.Tomoki@bp.MitsubishiElectric.co.jp;

Abstract

In 1997, Kani proved Kani's lemma, which asserts that a commutative diagram of four g -dimensional abelian varieties induces an isogeny between product abelian varieties of dimension $2g$, in counting the number of genus-2 curves admitting two distinct elliptic subcovers. In these years, Kani's lemma plays a fundamental role in isogeny-based cryptography: Kani's lemma has found numerous cryptographic applications, including both cryptanalysis and protocol construction. However, direct investigation into the lemma itself remains scarce.

In this work, we propose a generalization of Kani's lemma. We present a novel formulation that, given a commutative diagram of 2^{n+1} abelian varieties of dimension g , yields an isogeny of dimension $2^n g$. We further establish a connection between this generalized lemma and the theory of Clifford algebras, using the latter as a foundational tool in our construction. To exemplify our framework, we explicitly construct the resulting $2^n g$ -dimensional isogenies for the cases $n = 1, 2, 3$. The cases of $n = 2, 3$ provide nontrivial generalizations of the original Kani's lemma. This generalization is expected to have novel applications in the fields of both mathematics and cryptography.

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Statements and Declarations

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1 Introduction

Kani’s lemma, which appears in the proof of [1, Theorem 2.3], provides a construction of a $2g$ -dimensional isogeny from a commutative diagram of g -dimensional abelian varieties. Kani originally used this lemma to count genus-2 curves C that admit two fixed covers $C \rightarrow E$ and $C \rightarrow E'$ to given elliptic curves E and E' .

In recent years, the lemma has attracted considerable interest in the field of isogeny-based cryptography. Notably, in [2–4], it underlies novel key-recovery algorithms targeting SIDH (Supersingular Isogeny Diffie–Hellman) [5]. Additionally, several cryptographic protocols—e.g., SQISignHD [6] and FESTA [7]—have been proposed that employ Kani’s lemma as a core component in their design.

While there is abundant work on applications of Kani’s lemma, to the best of our knowledge, there have been no significant theoretical advancements concerning the lemma itself. In particular, Kani’s lemma currently applies only to a simple four-object commutative diagram of abelian varieties. Consequently, we pose the following natural research question:

Is there a generalization of Kani’s lemma that constructs isogenies from more intricate commutative diagrams?

In this work, we aim to answer this question by generalizing Kani’s lemma. This approach enables the construction of higher-dimensional isogenies and opens the door to potential new applications in both mathematics and cryptography.

1.1 Contributions

We first introduce the notion of an n -Kani relation, which is a map that yields an isogeny of $2^n g$ -dimensional abelian varieties constructed from a commutative diagram consisting of 2^{n+1} abelian varieties of dimension g . In this framework, the generalized Kani’s lemma asserts the existence of such an n -Kani relation.

We observe a connection between n -Kani relations and suitable Clifford algebras. Building on this insight, we prove the generalized Kani’s lemma using representations of Clifford algebras (Theorem 9). We also present examples for the 1-, 2-, and 3-Kani relations, which illustrate new nontrivial constructions of high-dimensional isogenies.

2 Preliminaries

In this section, we prepare background knowledge about abelian varieties and Clifford algebras, which are used to obtain our result for Kani’s lemma.

2.1 Abelian variety

In this subsection, we introduce some basic knowledge about abelian varieties. See [8, 9] for more details about abelian varieties.

Let K be a perfect field, and let A be an algebraic variety over K . We say A is an *abelian variety* over K if A has a group structure compatible with its algebraic structure. Let N be a positive integer. The N -torsion subgroup of A is the group defined by $A[N] := \{P \in A(\bar{K}) \mid NP = 0\}$, where $\bar{K}$ denotes an algebraic closure of

K . If K is a finite field and N is coprime to the characteristic of K , the group $A[N]$ is isomorphic to $(\mathbb{Z}/N\mathbb{Z})^{2 \dim A}$ as a group.

Let A and B be abelian varieties. A *homomorphism* between A and B is a morphism of algebraic varieties between A and B that is also a group morphism. We denote the group of homomorphisms between A and B by $\text{Hom}(A, B)$. An *endomorphism* of A is a homomorphism from A to A . The set of the endomorphisms of A has a ring structure. We denote this ring by $\text{End}(A)$. An *isogeny* $\phi: A \rightarrow B$ is a homomorphism such that ϕ is surjective and $\ker \phi$ is a finite subgroup.

If there is an isogeny between A and B , then it holds that $\dim A = \dim B$. Let $K(A)$ and $K(B)$ be function fields of A and B , respectively. There is a field morphism $\phi^*: K(B) \rightarrow K(A)$ defined by $f \mapsto f \circ \phi$. The *degree* of ϕ , denoted by $\deg \phi$, is the degree of the field extension $K(B)/\phi^*(K(A))$. For an integer d , we define the map $[d]_A$ by $[d]_A: A \rightarrow A; P \mapsto dP$.

We say ϕ is *separable* if the field extension $K(B)/\phi^*(K(A))$ is separable. For a separable isogeny ϕ , it holds that $\#\ker \phi = \deg \phi$. If there are separable isogenies $\phi: A \rightarrow B$ and $\phi': A \rightarrow B'$ with $\ker \phi = \ker \phi'$, then there is an isomorphism $\lambda: B \rightarrow B'$ satisfying $\phi' = \lambda \circ \phi$. Conversely, if we fix a finite subgroup G of A , then there is a separable isogeny mapping from A whose kernel is G .

Let $\phi: A \rightarrow B$ and $\phi': A \rightarrow B'$ be separable isogenies such that $\deg \phi$ and $\deg \phi'$ are coprime. The *pushforward of ϕ' by ϕ* is a separable isogeny mapping from B whose kernel is $\phi(\ker \phi')$. We denote by $[\phi]_* \phi'$ the pushforward of ϕ' by ϕ . We can assume that $([\phi']_* \phi) \circ \phi' = ([\phi]_* \phi') \circ \phi$ because $\ker(([\phi']_* \phi) \circ \phi') = \ker(([\phi]_* \phi') \circ \phi) = \ker \phi + \ker \phi'$. For an integer d with $\gcd(d, \deg \phi') = 1$, we have $[[d]_A]_* \phi' = \phi'$. Therefore, we define $[[d]_A]_* \phi' = \phi'$ for any integer d .

We denote the *dual abelian variety* of A by $\tilde{A}$. We do not give a definition of the dual abelian variety in this paper. See [9, Chapter 13] for the definition and construction of $\tilde{A}$. A *polarization* λ_A is an isogeny $\lambda_A: A \rightarrow \tilde{A}$, and a *polarized abelian variety* is a pair of an abelian variety A and its polarization. We say (A, λ_A) is a *principally polarized abelian variety* if the polarization λ_A is an isomorphism. In this paper, we only consider principally polarized abelian varieties, and often omit polarizations.

Let (A, λ_A) and (B, λ_B) be principally polarized abelian varieties. Let ϕ be a homomorphism $\phi: A \rightarrow B$. Then, we can construct the dual homomorphism $\tilde{\phi}: \tilde{B} \rightarrow \tilde{A}$ because the duality of abelian varieties is contravariant functorial. Since A and B are principally polarized, there is a homomorphism $\hat{\phi}: B \rightarrow A$ defined by $\hat{\phi} = \lambda_A^{-1} \circ \tilde{\phi} \circ \lambda_B$. If ϕ is an isogeny, the homomorphism $\hat{\phi}$ is also an isogeny, and we call this isogeny the *dual isogeny of ϕ* . Let d be a positive integer. If we have $\hat{\phi} \circ \phi = [d]_A$ and $\phi \circ \hat{\phi} = [d]_B$ for a homomorphism $\phi: A \rightarrow B$, then ϕ is an isogeny. We call ϕ a *d-isogeny*.

Let m_A, m_B be non-negative integers, let $A_0, \dots, A_{m_A}, B_0, \dots, B_{m_B}$ be principally polarized abelian varieties, and let $\Phi: \prod_{i=0}^{m_A} A_i \rightarrow \prod_{i=0}^{m_B} B_i$ be a homomorphism that can be represented by a matrix $M := (\phi_{i,j})_{i,j}$, where $\phi_{i,j}$ is a homomorphism from A_j to B_i . Then, the dual homomorphism $\hat{\Phi}$ can be represented by the matrix $M^* := (\hat{\phi}_{j,i})_{i,j}$.

We now state Kani's lemma (Proposition 1). We recall that a generalization of this lemma is the main goal of our paper.

Proposition 1 (Kani's lemma [1, 10]) *Let A , A_0 , A_1 , and A_{01} be principally polarized abelian varieties. We assume that there is a commutative diagram of separable isogenies:*

$$\begin{array}{ccc} A & \xrightarrow{\phi_0} & A_0 \\ \phi_1 \downarrow & & \downarrow [\phi_0]_* \phi_1 \\ A_1 & \xrightarrow{[\phi_1]_* \phi_0} & A_{01} \end{array}$$

Here, we also assume ϕ_0 (resp. ϕ_1) is a d_0 - (resp. d_1 -)isogeny with $\gcd(d_0, d_1) = 1$.

Then, there is a $(d_0 + d_1)$ -isogeny $\Phi: A_0 \times A_1 \rightarrow A \times A_{01}$ represented by

$$\Phi = \begin{pmatrix} \hat{\phi}_0 & \hat{\phi}_1 \\ -[\phi_0]_* \phi_1 & [\phi_1]_* \phi_0 \end{pmatrix}.$$

The kernel of Φ is given by

$$\left\langle \begin{pmatrix} \phi_0(P) \\ \phi_1(P) \end{pmatrix} \mid P \in A[d_0 + d_1] \right\rangle.$$

2.2 Clifford algebra

In this subsection, we briefly introduce a Clifford algebra and its properties related to our study. In particular, we give some already known results about representations of Clifford algebra $\mathbb{R}_{0,q}$. See [11] for more details about Clifford algebras.

Let K be a field, and let V be a K -linear space of dimension n . We assume that V is equipped with a quadratic form $Q: V \rightarrow K$. A *Clifford algebra* $\text{Cl}(V, Q)$ of (V, Q) is defined by a k -algebra

$$\text{Cl}(V, Q) := \left(\bigoplus_{i=0}^{\infty} V^{\otimes i} \right) / \langle v \otimes v - Q(v) \mid v \in V \rangle,$$

where $V^{\otimes i}$ denotes the tensor product of V with itself i times. We call a natural K -linear embedding $\iota: V \rightarrow \text{Cl}(V, Q)$ the *Clifford map*. If V is $\mathbb{R}^n$ (resp. $\mathbb{C}^n$) and a quadratic form Q has a diagonal norm

$$Q(x_1, \dots, x_n) = x_1^2 + \dots + x_p^2 - x_{p+1}^2 - \dots - x_n^2,$$

we denote the Clifford algebra of (V, Q) by $\mathbb{R}_{p,q}$ (resp. $\mathbb{C}_n$) where $q = n - p$.

We say a basis $\{e_1, \dots, e_n\}$ of (V, Q) is *orthogonal* if it holds that, in $\text{Cl}(V, Q)$,

$$e_i e_j = -e_j e_i \text{ for } i, j = 1, \dots, n \text{ and } i \neq j, \quad \text{and} \quad e_i^2 = Q(e_i).$$

The Clifford algebra $\text{Cl}(V, Q)$ is generated by $e_1, \dots, e_n$ as a K -algebra. Moreover, the Clifford algebra $\text{Cl}(V, Q)$ is 2^n -dimensional as a K -linear space. A basis of the linear space $\text{Cl}(V, Q)$ is given by

$$\{1, e_1, \dots, e_n, \dots, e_{i_1} \cdots e_{i_l}, \dots, e_1 \cdots e_n\},$$

where $i_1, \dots, i_l$ are indices with $1 \leq i_1 < \dots < i_l \leq n$.

The Clifford algebra $\text{Cl}(V, Q)$ has a universal property. Let A be a K -algebra with a K -linear map $\theta: V \rightarrow A$, and let 1_A be the multiplicative identity of A . Let $\{e_1, \dots, e_n\}$ be an orthogonal basis of (V, Q) . If the map θ satisfies $\theta(e_i)\theta(e_j) = -\theta(e_j)\theta(e_i)$ and $\theta(e_i)^2 = Q(e_i)1_A$ for $i, j = 1, \dots, n$ and $i \neq j$, then there is an algebra homomorphism $f: \text{Cl}(V, Q) \rightarrow A$ such that $f \circ \iota = \theta$, where $\iota: V \rightarrow \text{Cl}(V, Q)$ is the Clifford map. If

$$1_A, \theta(e_1), \dots, \theta(e_{i_1}) \cdots \theta(e_{i_l}), \dots, \theta(e_1) \cdots \theta(e_n)$$

are K -linearly independent, then f is injective. This implies that f is an embedding of $\text{Cl}(V, Q)$ in A .

2.2.1 Representation of $\mathbb{R}_{0,q}$

The classifications of $\mathbb{R}_{p,q}/\mathbb{C}_n$ are well known (*e.g.*, see [12]). We have the following isomorphisms between $\mathbb{R}_{p,q}$ and matrix algebras:

$$\mathbb{R}_{p,q} \cong \begin{cases} M_{2^{n/2}}(\mathbb{R}) & (\text{if } q - p \equiv 0, 6 \pmod{8}) \\ M_{2^{(n-1)/2}}(\mathbb{C}) & (\text{if } q - p \equiv 1, 5 \pmod{8}) \\ M_{2^{(n-2)/2}}(\mathbb{H}) & (\text{if } q - p \equiv 2, 4 \pmod{8}) \\ M_{2^{(n-3)/2}}(\mathbb{H}) \oplus M_{2^{(n-3)/2}}(\mathbb{H}) & (\text{if } q - p \equiv 3 \pmod{8}) \\ M_{2^{(n-1)/2}}(\mathbb{R}) \oplus M_{2^{(n-1)/2}}(\mathbb{R}) & (\text{if } q - p \equiv 7 \pmod{8}) \end{cases}.$$

We focus on representations of $\mathbb{R}_{0,q}$. From the above isomorphisms, if q can be represented by $q = 8a + 2^b - 1$ for $a \in \mathbb{Z}$ and $b = 0, 1, 2, 3$, then the irreducible representations of the Clifford algebra $\mathbb{R}_{0,q}$ are 2^{4a+b} -dimensional. In other words, the matrix ring $M_{2^s}(\mathbb{R})$ contains an image of $\mathbb{R}_{0,8a+2^b-1}$ under an irreducible representation, where $a = \lfloor s/4 \rfloor$ and $b = s - 4a$.

A representation of $\mathbb{R}_{0,8a+2^b-1}$ are explicitly given in [13, Proposition 3.5]. Here, we revisit the construction of the representation. We define the following three matrices:

$$L_1 := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad L_2 := \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad L_3 := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

We denote generators of the representation of $\mathbb{R}_{0,8a+2^b-1}$ by $M_1^{(a,b)}, \dots, M_{8a+2^b-1}^{(a,b)}$ in $M_s(\mathbb{R})$. We put $M_i^{(a,b)} := L_1^{\otimes 4a+b-i} \otimes L_2 \otimes I_{2^{i-1}}$ for $i = 1, \dots, 4a + b$, and define the other generators as follows:

- $M_i^{(a,0)} := L_1 \otimes M_{i-1}^{(a-1,3)}$ for $i = 4a + b + 1, \dots, 8a$,
- $M_i^{(a,1)} := L_1 \otimes M_{i-1}^{(a,0)}$ for $i = 4a + b + 1, \dots, 8a + 1$,
- $M_i^{(a,2)} := L_1 \otimes M_{i-1}^{(a,1)}$ for $i = 4a + b + 1, \dots, 8a + 2$, and

$$M_{8a+3}^{(a,2)} := L_3 \otimes L_2 \otimes L_3 \otimes I_{2^{4a-1}},$$

- $M_i^{(a,3)} := L_1 \otimes M_{i-1}^{(a,2)}$ for $i = 4a + b + 1, \dots, 8a + 4$,

$$M_{8a+5}^{(a,3)} := L_3 \otimes L_2 \otimes L_1 \otimes L_3 \otimes I_{2^{4a-1}}, \quad M_{8a+6}^{(a,3)} := L_3 \otimes I_2 \otimes L_2 \otimes L_3 \otimes I_{2^{4a-1}},$$

$$\text{and } M_{8a+7}^{(a,3)} := L_3 \otimes L_2 \otimes L_3 \otimes I_{2^{4a}}.$$

Here, we formally define $L_3 \otimes I_{2^{-1}}$ by 1. Moreover, we put $M_0^{(a,b)} = I_{2^{4a+b}}$. We note that all matrices $M_i^{(a,b)}$ are defined over $\{-1, 0, 1\}$.

In the above construction, the matrices $M_1^{(a,b)}, \dots, M_{4a+b}^{(a,b)}$ play an important role among the generators $M_1^{(a,b)}, \dots, M_{8a+2b-1}^{(a,b)}$. Specifically, if we ignore the signs of generators or, in other words, consider reductions to matrices over $\mathbb{F}_2$, then $M_1^{(a,b)}, \dots, M_{4a+b}^{(a,b)}$ can generate the other generators.

Lemma 2 *Let $\overline{M}_i^{(a,b)}$ be the reduction of $M_i^{(a,b)}$ over $\mathbb{F}_2$. Then, it holds that the matrices $\overline{M}_1^{(a,b)}, \dots, \overline{M}_{4a+b}^{(a,b)}$ can generate $\overline{M}_i^{(a,b)}$ for any $i = 1, \dots, 8a + 2b - 1$. In particular, we have, for $i > 4a + b$,*

$$\overline{M}_i^{(a,b)} = \begin{cases} \overline{M}_{i-4a-b-1}^{(a,b)} \cdot \overline{M}_{i-4a-b}^{(a,b)} \cdot \overline{M}_{i-4a-b+1}^{(a,b)} & (\text{if } i \equiv b+1 \pmod{4}) \\ \overline{M}_{i-4a-b-2}^{(a,b)} \cdot \overline{M}_{i-4a-b}^{(a,b)} \cdot \overline{M}_{i-4a-b+1}^{(a,b)} & (\text{if } i \equiv b+2 \pmod{4}) \\ \overline{M}_{i-4a-b-3}^{(a,b)} \cdot \overline{M}_{i-4a-b-2}^{(a,b)} \cdot \overline{M}_{i-4a-b}^{(a,b)} & (\text{if } i \equiv b+3 \pmod{4}) \\ \overline{M}_{i-4a-b-3}^{(a,b)} \cdot \overline{M}_{i-4a-b-2}^{(a,b)} \cdot \overline{M}_{i-4a-b-1}^{(a,b)} & (\text{if } i \equiv b+4 \pmod{4}) \end{cases}.$$

Proof It is easy to see that $\overline{M}_i^{(a,b)}$ can be generated by $\overline{M}_1^{(a,b)}, \dots, \overline{M}_{4a+b}^{(a,b)}$, because we have

$$M_j^{(a,b)} \equiv I_{2^{4a+b-j}} \otimes L_2 \otimes I_{2^{j-1}} \pmod{2} \quad (1)$$

for $j = 1, \dots, 4a + b$.

From the construction of $M_i^{(a,b)}$, if we have $i > 4a + b$ and $i - 4a - b \equiv 1 \pmod{4}$, then it holds that

$$M_i^{(a,b)} = L_1 \otimes \dots \otimes L_1 \otimes L_3 \otimes L_2 \otimes L_3 \otimes I_2 \otimes \dots \otimes I_2.$$

Moreover, if we denote the number of L_1 appearing at $M_i^{(a,b)}$ by m_i for $i > 4a + b$ and $i - 4a - b \equiv 1 \pmod{4}$, then m_i satisfy the following properties:

$$m_{i-4} = m_i + 4,$$

$$\begin{cases} m_{8a-3} = 2 & (\text{if } b = 0) \\ m_{8a-2} = 3 & (\text{if } b = 1) \\ m_{8a+3} = 0 & (\text{if } b = 2) \\ m_{8a+4} = 1 & (\text{if } b = 3) \end{cases}.$$

Therefore, we have $m_i = 8a + 2b - 1 - i$. From the same discussions, we have

$$M_i^{(a,b)} = \begin{cases} L_1^{\otimes(8a+2b-1-i)} \otimes L_3 \otimes L_2 \otimes L_3 \otimes I_2 \otimes \dots \otimes I_2 & (\text{if } i \equiv b+1 \pmod{4}) \\ L_1^{\otimes(8a+2b-1-i)} \otimes L_3 \otimes L_2 \otimes L_1 \otimes L_3 \otimes I_2 \otimes \dots \otimes I_2 & (\text{if } i \equiv b+2 \pmod{4}) \\ L_1^{\otimes(8a+2b-i)} \otimes L_3 \otimes I_2 \otimes L_2 \otimes L_3 \otimes I_2 \otimes \dots \otimes I_2 & (\text{if } i \equiv b+3 \pmod{4}) \\ L_1^{\otimes(8a+2b+1-i)} \otimes L_3 \otimes L_2 \otimes L_3 \otimes I_2 \otimes \dots \otimes I_2 & (\text{if } i \equiv b+4 \pmod{4}) \end{cases}.$$

Since $L_1 \equiv I_2 \pmod{2}$, it is easy to check the assertion from the above equations and Equation (1). $\square$

Remark 1 An irreducible representation of $\mathbb{R}_{0,8a+2^b-1}$ is not unique. Lemma 2 shows one of the constructions of such representations.

Another significant property of the matrices $M_1^{(a,b)}, \dots, M_{4a+b}^{(a,b)}$ is that they give an embedding of $\mathbb{R}_{0,4a+b} \subset \mathbb{R}_{0,8a+2^b-1}$ in $M_{2^{4a+b}}(\mathbb{R})$. We show this property in Lemma 3.

Lemma 3 *Let $\{e_1, \dots, e_{4a+b}\}$ be an orthogonal basis of $\mathbb{R}_{0,4a+b}$. We put $e_\emptyset = 1$, and for a set $S \subset \{1, \dots, 4a+b\}$, we put $e_S := \prod_{i \in S} e_i$. Then, the matrices $M_1^{(a,b)}, \dots, M_{4a+b}^{(a,b)}$ give an embedding of $\mathbb{R}_{0,4a+b}$ in $M_{2^{4a+b}}(\mathbb{R})$ by*

$$\begin{aligned} \mathbb{R}_{0,4a+b} &\longrightarrow M_{2^{4a+b}}(\mathbb{R}) \\ \sum_{S \subset \{1, \dots, 4a+b\}} a_S e_S &\longmapsto \sum_{S \subset \{1, \dots, 4a+b\}} a_S \cdot \prod_{i \in S} M_i^{(a,b)}, \end{aligned}$$

where $a_S \in \mathbb{R}$ for $S \subset \{1, \dots, 4a+b\}$, and we formally let $\prod_{i \in \emptyset} M_i^{(a,b)} = I_{2^{4a+b}}$.

Proof It follows from Equation (1) that

$$\prod_{i \in S} M_i^{(a,b)} \equiv \bigotimes_{j=1}^{4a+b} \tilde{L}_j \pmod{2} \quad \text{with} \quad \tilde{L}_j = \begin{cases} \tilde{L}_j = I_2 & (\text{if } j \notin S) \\ \tilde{L}_j = L_2 & (\text{if } j \in S) \end{cases}.$$

Therefore, the matrices $\prod_{i \in S} M_i^{(a,b)}$ for $S \subset \{1, \dots, 4a+b\}$ are $\mathbb{R}$ -linearly independent. From the universal property of the Clifford algebra, the proof of the assertion holds. $\square$

3 Kani isogeny

In this section, we introduce several mathematical concepts and notations, which are used in our generalization of Kani's lemma.

3.1 Introduction to n -Kani diagrams

In this subsection, we introduce some concepts related to an n -Kani diagram, which is a generalization of an isogeny diamond, or cryptographically, a SIDH diagram. Specifically, a 1-Kani diagram is exactly an isogeny diamond.

We first define an n -isogeny diamond configuration. An n -isogeny diamond configuration is a natural extension of an isogeny diamond configuration introduced in [1], and gives our starting line to consider an extension of Kani's lemma.

Definition 1 (n -isogeny diamond configuration) Let A, B be principally polarized abelian varieties, and let $\phi: A \rightarrow B$ be a separable isogeny between A and B . Let $d_0, \dots, d_n$ be pairwise coprime positive integers, and let $H_0, \dots, H_n$ be finite subgroups of A satisfying

- H_i is a maximal isotropic subgroup of $A[d_i]$ with $H_i \cong (\mathbb{Z}/d_i\mathbb{Z})^{\dim A}$,

- $H_i \subset \ker \phi$,
- $\prod_{i=0}^n \#H_i = \deg f$.
- $H_i \cap H_j = \emptyset$ for all $i, j = 0, \dots, n$ with $i \neq j$.

We say $(A, \phi, H_0, \dots, H_n)$ an n -isogeny diamond configuration of order $(d_0, \dots, d_n)$.

From the same discussion as in [1, Remark 2.2], we find a relationship between an n -isogeny diamond configuration and a commutative diagram forming an $(n+1)$ -dimensional hypercube. That is, from $(A, \phi, H_0, \dots, H_n)$, we can construct isogenies $\phi_i: A \rightarrow A/H_i$ for $i = 0, \dots, n$, and obtain a commutative diagram in the shape of an $(n+1)$ -dimensional hypercube generated from ϕ_i 's and their pushforwards.

Let a tuple $\mathcal{D} := (A, \phi, H_0, \dots, H_n)$ be an n -isogeny diamond configuration, and let S, S' be finite subsets of $\{0, \dots, n\}$. We denote by A_S , a principally polarized abelian variety that is a codomain variety of a d_S -isogeny

$$\phi_S: A \longrightarrow A_S \quad \text{with} \quad \ker \phi_S = \sum_{i \in S} H_i,$$

where $d_S = \prod_{i \in S} d_i$. It is easy to see that $\{A_S \mid S \subset \{0, \dots, n\}\}$ is the set of varieties appearing in the diagram in the shape of an n -dimensional hypercube obtained from $\mathcal{D}$. In particular, we have $A_\emptyset = A$ and $A_{\{0, \dots, n\}} = B$. We denote by ϕ_S , a d_S -isogeny mapping from $A_\emptyset$ to A_S , and denote by $S'_* \phi_S$, an isogeny $[\phi_{S'}]_* \phi_S$. Here, we let $\phi_\emptyset$ be the identity map of $A_\emptyset$, and let $S_* \phi_\emptyset$ be the identity map of A_S . Also, although it is an abuse of notation, we denote the dual isogeny of $S'_* \phi_S$ by $S'_* \hat{\phi}_S$. Under these notations, there is a commutative diagram:

$$\begin{array}{ccc} A_{S \cap S'} & \xrightarrow{(S \cap S')_* \phi_{S \setminus S'}} & A_S \\ (S \cap S')_* \phi_{S' \setminus S} \downarrow & & \downarrow S_* \phi_{S' \setminus S} \\ A_{S'} & \xrightarrow{S'_* \phi_{S \setminus S'}} & A_{S \cup S'} \end{array}$$

Now, we explicitly write down the definition of an n -Kani diagram, which is the n -dimensional hypercubic diagram described in the previous paragraph.

Definition 2 (n -Kani diagram) An n -Kani diagram over $(A_S)_{S \subset \{0, \dots, n\}}$ of order $(d_0, \dots, d_n)$ is the commutative diagram in the shape of an $(n+1)$ -dimensional hypercube that consists of A_S and $S_* \phi_{\{i\}}$'s, which is derived from an n -dimensional isogeny diamond configuration of order $(d_0, \dots, d_n)$. We denote it by $D_{(\phi_{\{0\}}, \dots, \phi_{\{n\}})}$.

Let $D_{(\phi_{\{0\}}, \dots, \phi_{\{n\}})}$ and $D_{(\phi'_{\{0\}}, \dots, \phi'_{\{n\}})}$ be n -Kani diagrams over $(A_S)_{S \subset \{0, \dots, n\}}$ and $(A'_S)_{S \subset \{0, \dots, n\}}$ respectively. We say diagrams $D_{(\phi_{\{0\}}, \dots, \phi_{\{n\}})}$ and $D_{(\phi'_{\{0\}}, \dots, \phi'_{\{n\}})}$ are equivalent if it holds that, for all $S \subset \{0, \dots, n\}$, there are isomorphisms $\lambda_S: A_S \rightarrow A'_S$ with $S_* \phi'_{\{i\}} \circ \lambda_S = \lambda_{S \cup \{i\}} \circ S_* \phi_{\{i\}}$ for all $i = 0, \dots, n$.

Let $\beta_0, \dots, \beta_n$ be integers. For an n -Kani diagram $D_{(\phi_{\{0\}}, \dots, \phi_{\{n\}})}$, we can construct another commutative diagram that is generated by $\beta_0 \phi_{\{0\}}, \dots, \beta_n \phi_{\{n\}}$. We denote this commutative diagram by $D_{(\beta_0 \phi_{\{0\}}, \dots, \beta_n \phi_{\{n\}})}$. We denote the set of isomorphism classes of such commutative diagrams over $(A_S)_{S \subset \{0, \dots, n\}}$, by $\mathcal{D}_{(A_S)_{S \subset \{0, \dots, n\}}}$. We call $\mathcal{D}_{(A_S)_{S \subset \{0, \dots, n\}}}$ an n -Kani set over $(A_S)_{S \subset \{0, \dots, n\}}$.

3.2 Definition of Kani isogenies

In this subsection, we provide the definition of a *Kani isogeny*, which is a generalization of the isogeny derived from Kani's lemma. We can say that Kani's lemma is a lemma that gives an embedding from an n -Kani set to a set of Kani isogenies, which we will call an n -Kani relation.

Definition 3 (Kani isogeny) For $i = 1, \dots, m$, let $A_{i, dom}$ and $A_{i, codom}$ be principally polarized abelian varieties with

$$\dim A_{1, dom} = \dots = \dim A_{m, dom} = \dim A_{1, codom} = \dots = \dim A_{m, codom}.$$

We define A_{dom} and A_{codom} by

$$A_{dom} := \prod_{i=1}^m A_{i, dom}, \quad A_{codom} := \prod_{i=1}^m A_{i, codom}.$$

A Kani isogeny of order N for (A_{dom}, A_{codom}) is an N -isogeny

$$\Phi: A_{dom} \longrightarrow A_{codom}$$

such that it can be represented by a matrix $M_\Phi = (\phi_{i,j})_{i,j=1, \dots, m}$, where $\phi_{i,j}$ belongs to $\text{Hom}(A_{j, dom}, A_{i, codom})$.

Remark 2 Recall that the dual isogeny $\hat{\Phi}$ of Φ is represented by the adjoint matrix $M_\Phi^* = (\hat{\phi}_{j,i})_{i,j}$ of M_Φ . Therefore, the isogeny Φ is a N -isogeny if and only if $M_\Phi^* \cdot M_\Phi = [N]_{A_{dom}}$ and $M_\Phi \cdot M_\Phi^* = [N]_{A_{codom}}$.

Remark 3 In the paper [1], isogenies derived from Kani's lemma are described as isogenies whose kernel are non-diagonal, reducible, and maximally isotropic subgroups of $A_{dom}[N]$. A Kani isogeny is actually an isogeny whose kernel is a reducible and maximally isotropic subgroup of $A_{dom}[N]$. Here, we omit “non-diagonal” from the definition, because it is more suitable as far as we focus on a construction of high-dimensional isogenies. For example, a definition with “non-diagonal” cannot handle the case of $N = 1$.

We now introduce a novel equivalence relation among isogenies between two products of abelian varieties. Assume that there are two isogenies

$$\Phi_A: \prod_{i=0}^m A_i \longrightarrow \prod_{i=0}^m A'_i, \quad \Phi_B: \prod_{i=0}^m B_i \longrightarrow \prod_{i=0}^m B'_i,$$

where A_i , B_i , A'_i , and B'_i are principally polarized abelian varieties that have the common dimension. We define an equivalence relation $\sim_d$ between Φ_A and Φ_B as we say $\Phi_A \sim_d \Phi_B$ if there are isomorphisms

$$\lambda_i: A_i \rightarrow B_i \quad \text{and} \quad \lambda'_i: A'_i \rightarrow B'_i \quad \text{for } i = 0, \dots, m$$

such that

$$\text{diag}(\lambda'_0, \dots, \lambda'_m) \circ \Phi_A = \Phi_B \circ \text{diag}(\lambda_0, \dots, \lambda_m).$$

We denote by $\mathcal{I}_{(A_{dom}, A_{codom})}$, the set of equivalent classes with respect to $\sim_d$ of Kani isogenies for (A_{dom}, A_{codom}) . It is easy to see that the set $\mathcal{I}_{(A_{dom}, A_{codom})}$ can be embedded in $\text{Hom}(A_{dom}, A_{codom}) / \sim_d$.

Definition 4 (n -Kani relation) Let S be a subset of $\{0, \dots, n\}$, and let A_S be a principally polarized abelian variety. Let $\mathcal{S}_{dom}$ and $\mathcal{S}_{codom}$ be sets of some subsets of $\{0, \dots, n\}$ with $\#\mathcal{S}_{dom} = \#\mathcal{S}_{codom}$ and $\mathcal{S}_{dom} \cap \mathcal{S}_{codom} = \emptyset$. We define A_{dom} and A_{codom} by

$$A_{dom} := \prod_{S \in \mathcal{S}_{dom}} A_S \quad \text{and} \quad A_{codom} := \prod_{S \in \mathcal{S}_{codom}} A_S.$$

Let f be a polynomial in $\mathbb{Z}[x_0, \dots, x_n]$.

An n -Kani relation over $(A_S)_{S \subset \{0, \dots, n\}}$ to (A_{dom}, A_{codom}) associated with f is an injective map

$$\begin{aligned} R_{(A_S)_{S \subset \{0, \dots, n\}}} : \mathcal{D}_{(A_S)_{S \subset \{0, \dots, n\}}} &\longrightarrow \mathcal{I}_{(A_{dom}, A_{codom})} \\ D_{(\phi_{\{0\}}, \dots, \phi_{\{n\}})} &\longmapsto (r_{S, S'}((\phi_{\{i\}})_{i \in S \ominus S'}))_{(S, S') \in A_{codom} \times A_{dom}} \end{aligned}$$

such that an n -Kani diagram of order $(d_0, \dots, d_n)$ maps to a Kani isogeny of order $f(d_0, \dots, d_n)$, and $r_{S, S'}$ is a map satisfying

$$r_{S, S'}(D_{(\beta_0 \phi_{\{0\}}, \dots, \beta_n \phi_{\{n\}})}) = \left(\prod_{i \in S \ominus S'} \beta_i \right) \cdot r_{S, S'}(D_{(\phi_{\{0\}}, \dots, \phi_{\{n\}})}),$$

where $S \ominus S'$ denotes the symmetric difference of S and S' (i.e., $S \ominus S' = (S \cup S') \setminus (S \cap S')$).

4 Generalizations of Kani's lemma

In this section, we provide our main result for generating Kani's lemma using Clifford algebras. In Section 4.1, we show the reason why Clifford algebras are applicable to obtain a generalization of Kani's lemma. Section 4.2 gives our main theorem (Theorem 9), which provides a construction of an n -Kani relation. Section 4.3 shows explicit constructions of 1, 2, 3-Kani relations. Finally, we explain a limitation for extending Kani's lemma based on our approach in Section 4.4.

4.1 Connection with Clifford algebras and n -Kani relations

In this subsection, we see the relationship between an n -Kani relation and a representation of the Clifford algebra $\mathbb{R}_{0,n}$ in the matrix ring $M_{2^{m+1}}(\mathbb{R})$. This gives an intuition that we could construct an n -Kani relation from a representation of $\mathbb{R}_{0,n}$.

Proposition 4 shows that we can construct a representation of $\mathbb{R}_{0,n}$ if there is a specific n -Kani relation.

Proposition 4 Let A_S for $S \subset \{0, \dots, n\}$ be principally polarized abelian varieties, let m be a positive integer with $m \leq 2^n$, and let $\mathcal{S}_{dom}$ and $\mathcal{S}_{codom}$ be sets consisting of subsets of $\{0, \dots, n\}$ with $\#\mathcal{S}_{dom} = \#\mathcal{S}_{codom} = m$. We define $A_{dom} := \prod_{S \in \mathcal{S}_{dom}} A_S$ and $A_{codom} := \prod_{S' \in \mathcal{S}_{codom}} A_{S'}$. Let R be an n -Kani relation over $(A_S)_{S \subset \{0, \dots, n\}}$ to (A_{dom}, A_{codom}) associated with a polynomial f .

If the polynomial f is $f := \alpha_0 x_0 + \dots + \alpha_n x_n$ with $\alpha_0, \dots, \alpha_n \in \mathbb{Z}_{>0}$, then we can construct an algebra homomorphism from the Clifford algebra $\mathbb{R}_{0,n}$ to $\mathbb{R} \otimes \text{End}(A_{dom})$.

Proof We first show that R is a group morphism. It suffices to prove that $r_{S,S'} = 0$ for any $S' \in \mathcal{S}_{dom}$ and $S \in \mathcal{S}_{codom}$ with $\#(S \ominus S') > 1$, where $r_{S,S'}$ is a multilinear map $r_{S,S'}: \mathcal{D}_{(A_S)_{S \subset \{0, \dots, n\}}} \rightarrow \text{Hom}(A_{S'}, A_S)$ related to the (S, S') -th entry of R .

We suppose that $r_{S,S'} \neq 0$ for some $(S', S) \in \mathcal{S}_{dom} \times \mathcal{S}_{codom}$ with $\#(S \ominus S') > 1$. We fix an n -Kani diagram $D_{(\phi_{\{0\}}, \dots, \phi_{\{n\}})}$ over $(A_S)_{S \subset \{0, \dots, n\}}$ of order $(d_0, \dots, d_n)$ with $d_i \neq 0$ for all $i = 0, \dots, n$. We define

$$R_d: \begin{array}{ccc} \mathbb{Z}^{n+1} & \longrightarrow & \mathbb{Z} \cong \mathbb{Z} \cdot I_{2^{m+1}} \\ (\beta_0, \dots, \beta_n) & \longmapsto & M^* M \end{array},$$

where $M = R(D_{(\beta_0 \phi_{\{0\}}, \dots, \beta_n \phi_{\{n\}})})$. From the definition of R , the map R_d is represented by a polynomial

$$R_d(y_0, \dots, y_n) = \sum_{i=0}^n \alpha_i d_i \cdot y_i^2. \quad (2)$$

By focusing on $r_{S,S'}$, we can find the term $\prod_{i \in S \ominus S'} y_i^2$ in R_d . Since the other entries in the S' -th column in R do not provide the term $\prod_{i \in S \ominus S'} y_i^2$, this term does not vanish. This contradicts Equation (2) because it does not have the term $\prod_{i \in S \ominus S'} y_i^2$. Hence, we conclude that $r_{S,S'} = 0$, and R is a linear map.

Put $M'_i := R(D_{(0, \dots, 0, \phi_{\{i\}}, 0, \dots, 0)})$. Then, for $i, j = 0, \dots, n$ and $i \neq j$, we have

$$(M'_i + M'_j)^* \cdot (M'_i + M'_j) = (\alpha_i d_i + \alpha_j d_j) I_{2^{m+1}} + M_i'^* M_j' + M_j'^* M_i'.$$

Since $M'_i + M'_j = R(D_{(0, \dots, 0, \phi_{\{i\}}, 0, \dots, 0)} + D_{(0, \dots, 0, \phi_{\{j\}}, 0, \dots, 0)})$, it holds that

$$(M'_i + M'_j)^* \cdot (M'_i + M'_j) = (\alpha_i d_i + \alpha_j d_j) I_{2^{m+1}}.$$

This implies $M_i'^* M_j' + M_j'^* M_i' = 0$.

We put $M_i := M_0'^* M_i' \in \text{End}(A_{dom})$ for $i = 0, \dots, n$. Then, we have, for $i, j = 0, \dots, n$ and $i \neq j$,

$$\begin{aligned} M_i M_j + M_j M_i &= (M_0'^* M_i')(M_0'^* M_j') + (M_0'^* M_j')(M_0'^* M_i') \\ &= -(M_i'^* M_0')(M_0'^* M_j') - (M_j'^* M_0')(M_0'^* M_i') \\ &= -\alpha_0 d_0 (M_i'^* M_j' + M_j'^* M_i') = 0. \end{aligned}$$

Moreover, we have, for $i = 1, \dots, n$,

$$M_i^2 = (M_0'^* M_i')^2 = -(M_i'^* M_0')(M_0'^* M_i') = -\alpha_0 d_0 M_i'^* M_i' = -\alpha_0 \alpha_i d_0 d_i I_{2^{m+1}},$$

and $M_0^2 = \alpha_0^2 d_0^2 I_{2^{m+1}}$. Therefore, from the universal property of Clifford algebras, we conclude that there is an algebra homomorphism from $\mathbb{R}_{0,n}$ to $\mathbb{R} \otimes \text{End}(A_{dom})$ by $e_i \mapsto \frac{1}{\sqrt{\alpha_0 \alpha_i d_0 d_i}} M_i$, where $\{e_1, \dots, e_n\}$ is an orthogonal basis of $\mathbb{R}_{0,n}$. $\square$

We immediately obtain the following corollary from this proposition.

Corollary 5 *Let A be a principally polarized abelian variety, let m be a positive integer with $m \leq 2^n$. Let R be an n -Kani relation over $(A)_{S \subset \{0, \dots, n\}}$ to (A^{m+1}, A^{m+1}) associated with a polynomial f .*

If f is the polynomial $f := \alpha_0 x_0 + \dots + \alpha_n x_n$ with $\alpha_0, \dots, \alpha_n \in \mathbb{Z}_{>0}$, then we have a representation of the Clifford algebra $\mathbb{R}_{0,n}$ in $M_{2^{m+1}}(\mathbb{R})$.

Proof Note that $M_{2^{m+1}}(\mathbb{Z}) \subset \text{End}(A^{m+1})$ by considering homomorphisms represented by scalar matrices. We focus on n -Kani diagrams $D_{(1,0,\dots,0)}, \dots, D_{(0,\dots,0,1)}$ over $(A)_{S \subset \{0, \dots, n\}}$. Then, from the same discussion in the proof of Proposition 4, we can construct an algebra homomorphism from $\mathbb{R}_{0,n}$ to $\mathbb{R} \otimes M_{2^{m+1}}(\mathbb{Z}) \cong M_{2^{m+1}}(\mathbb{R})$. $\square$

From the above proposition and corollary, we find that an n -Kani relation associated with a linear polynomial gives a representation of $\mathbb{R}_{0,n}$ in a matrix ring. Therefore, it should be a sound approach to construct an n -Kani relation from a representation of the Clifford algebra $\mathbb{R}_{p,q}$ with $\mathbb{R}_{0,n} \subset \mathbb{R}_{p,q}$. We explicitly give such constructions in Section 4.2.

Remark 4 It is an open problem whether a general n -Kani relation leads a representation of $\mathbb{R}_{0,n}$, or not.

4.2 Explicit construction of n -Kani relations

In this subsection, we provide an explicit construction of an n -Kani relation. This construction proves a generalization of Kani's lemma (Theorem 9).

Let a, b be positive integers with $n = 4a + b$ and $b = 0, 1, 2, 3$. We have $2^n \times 2^n$ -matrices $M_1^{(a,b)}, \dots, M_{8a+2^b-1}^{(a,b)}$ from Section 2.2.1. These matrices give a representation of $\mathbb{R}_{0,8a+2^b-1}$ in $M_{2^n}(\mathbb{R})$. Moreover, the matrices $M_1^{(a,b)}, \dots, M_n^{(a,b)}$ can construct an embedding of $\mathbb{R}_{0,n}$ in $M_{2^n}(\mathbb{R})$ from Lemma 3. We often denote $M_i^{(a,b)}$ by $M_i^{(n)}$.

Our goal is to have an embedding of $\mathcal{D}_{(A_S)_{S \subset \{0, \dots, n\}}}$ in $\mathcal{I}(A_{\text{dom}}, A_{\text{codom}})$. Thus, the entries of the matrix are constrained by A_{dom} and A_{codom} . For example, the $(1, 1)$ -th entry of the matrices should be a homomorphism from $A_{1, \text{dom}}$ to $A_{1, \text{codom}}$. Therefore, we could need to permute the indices of the matrices derived from an embedding of $\mathbb{R}_{0,n}$ into $M_{2^n}(\mathbb{R})$. The following lemma and its proof provide an appropriate way to decide the indices.

Lemma 6 *Let $\mathcal{S}_{\text{odd}}^{(n)}$ be a set defined by*

$$\mathcal{S}_{\text{odd}}^{(n)} = \{S \mid S \subset \{0, \dots, n\}, \#S: \text{odd}\}.$$

We put $M_i^{(n)} = (\epsilon_{i,k,l}^{(n)})_{k,l}$ for $i = 1, \dots, n$. Then, there is a map η_n mapping from $\mathcal{S}_{\text{odd}}^{(n)}$ to $\{1, \dots, 2^n\}$ such that

$$\epsilon_{i, \eta(S), \eta(S')}^{(n)} = \begin{cases} \pm 1 & (\text{if } S \ominus S' = \{0, i\}) \\ 0 & (\text{otherwise}) \end{cases},$$

where $S \ominus S'$ denotes the symmetric difference of S and S' .

Proof We denote by $\mathcal{S}^{(n)}$, the set of the subsets of $\{0, \dots, n\}$. We define a bijection η by

$$\eta: \bigcup_{n=0}^{\infty} \mathcal{S}^{(n)} \longrightarrow \mathbb{Z}_{\geq 0}$$

$$S \longmapsto \sum_{i \in S} 2^i,$$

where we put $\eta(\emptyset) = 0$. Since we have $\eta(\mathcal{S}^{(n)}) = \{0, \dots, 2^{n+1} - 1\}$, the set $\eta(\mathcal{S}_{\text{odd}}^{(n)})$ is a subset of $\{0, \dots, 2^{n+1} - 1\}$ whose cardinality is 2^n . In particular, we can define a bijection $\eta_n: \mathcal{S}_{\text{odd}}^{(n)} \rightarrow \{1, \dots, 2^n\}$. By swapping the images of η_n appropriately, we can assume that $\eta_n(S) < \eta_n(S')$ if and only if $\eta(S) < \eta(S')$.

We show that the map η_n satisfies the property in the statement of Lemma 6, via mathematical induction. We recall that $M_i^{(n)} = L_1 \otimes M_i^{(n-1)}$ for $i = 1, \dots, n-1$ and $M_n^{(n)} = L_2 \otimes I_{2^{n-1}}$, where $L_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $L_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

If $n = 1$, then the assertion follows from

$$\eta_1(\{0\}) = 1, \quad \eta_1(\{1\}) = 2, \quad M_1^{(1)} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

We now introduce several notations for the rest of the proof. For $m = 1, \dots, n$, let $\mathcal{S}_{\text{even}}^{(m)}$ be the set $\{T \mid T \in \mathcal{S}^{(m)}, \#T: \text{ even}\}$. We can also define a bijection $\eta'_m: \mathcal{S}_{\text{even}}^{(m)} \rightarrow \{1, \dots, 2^m\}$ by $\eta'_m(T) := \eta_m(T \ominus \{0\})$. It follows from the definition of η that $\eta(T \ominus \{0\}) = \eta(T) \pm 1$. If $T, T' \in \mathcal{S}_{\text{even}}^{(m)}$ and $T \neq T'$, then $|\eta(T) - \eta(T')| > 1$. Therefore, it holds that $\eta(T \ominus \{0\}) > \eta(T' \ominus \{0\})$ if and only if $\eta(T) > \eta(T')$. This implies $\eta'_m(T) > \eta'_m(T')$ if and only if $\eta(T) > \eta(T')$. Note that we have

$$\mathcal{S}_{\text{odd}}^{(m)} = \mathcal{S}_{\text{odd}}^{(m-1)} \sqcup \{T \cup \{m\} \mid T \in \mathcal{S}_{\text{even}}^{(m-1)}\},$$

where the symbol $\sqcup$ denotes a disjoint union. Therefore, it holds that

$$\eta_m(S) = \begin{cases} \eta_{m-1}(S) & (\text{if } S \in \mathcal{S}_{\text{odd}}^{(m-1)}) \\ \eta'_{m-1}(S \setminus \{m\}) + 2^{m-1} & (\text{if } S \in \mathcal{S}_{\text{odd}}^{(m)} \setminus \mathcal{S}_{\text{odd}}^{(m-1)}) \end{cases}.$$

From now on, we assume that the assertion holds for η_{n-1} . We first focus on the case that $S \ominus S' = \{0, i\}$ for $i \neq n$. Note that, in this case, S and S' are both contained in $\mathcal{S}_{\text{odd}}^{(n-1)}$ or $\{T \cup \{n\} \mid T \in \mathcal{S}_{\text{even}}^{(n-1)}\}$. This shows that $S \ominus S' = \{0, i\}$ if and only if there are $\tilde{S}, \tilde{S}' \in \mathcal{S}_{\text{odd}}^{(n-1)}$ such that

$$(\eta_n(S), \eta_n(S')) = (\eta_{n-1}(\tilde{S}), \eta_{n-1}(\tilde{S}')) \quad \text{or} \quad (\eta_{n-1}(\tilde{S}) + 2^{n-1}, \eta_{n-1}(\tilde{S}') + 2^{n-1}).$$

Hence, if $i \neq n$, the assertion follows from the relation $M_i^{(n)} = L_1 \otimes M_i^{(n-1)}$ and the assumption of the induction.

We finally consider the case that $S \ominus S' = \{0, n\}$. In this case, either S or S' — but not both — is contained in $\mathcal{S}_{\text{odd}}^{(n-1)}$. Therefore, we have $S \ominus S' = \{0, n\}$ if and only if

$$S = (S' \cup \{n\}) \ominus \{0\} \quad (\text{if } S' \in \mathcal{S}_{\text{odd}}^{(n-1)}),$$

$$\text{or} \quad S' = (S \cup \{n\}) \ominus \{0\} \quad (\text{if } S \in \mathcal{S}_{\text{odd}}^{(n-1)}).$$

Hence, it holds that $S \ominus S' = \{0, n\}$ if and only if

$$\eta_n(S) = \eta_n(S') + 2^{n-1} \quad (\text{if } S' \in \mathcal{S}_{\text{odd}}^{(n-1)}),$$

$$\text{or} \quad \eta_n(S') = \eta_n(S) + 2^{n-1} \quad (\text{if } S \in \mathcal{S}_{\text{odd}}^{(n-1)}).$$

Since $M_n^{(n)} = L_2 \otimes I_{2^{n-1}}$, this completes the rest part of the proof. $\square$

We define $\eta'_n: \mathcal{S}_{even}^{(n)} \rightarrow \{1, \dots, n\}$ as in the proof of Lemma 6. Let S_k (resp. S'_k) be a set in $\mathcal{S}_{odd}^{(n)}$ (resp. $\mathcal{S}_{even}^{(n)}$) such that $k = \eta_n(S_k)$ (resp. $k = \eta'_n(S'_k)$). We define $(A_S)_{S \subset \{0, \dots, n\}}$ in the same way as in Sections 3.1 and 3.2. We also define A_{odd} and A_{even} by $A_{odd} := \prod_{S \in \mathcal{S}_{odd}^{(n)}} A_S$ and $A_{even} := \prod_{S' \in \mathcal{S}_{even}^{(n)}} A_{S'}$, respectively. We define principally polarized abelian varieties $A_1, \dots, A_{2^n}$ by $A_k := A_{S_k}$ for $S_k \in \mathcal{S}_{odd}^{(n)}$, and we also define $A'_1, \dots, A'_{2^n}$ by $A'_k := A_{S'_k}$ for $S'_k \in \mathcal{S}_{even}^{(n)}$. We let $A_{odd} = \prod_{j=1}^{2^n} A_j$ and $A_{even} = \prod_{j=1}^{2^n} A'_j$ by changing the indices.

Let $\mathcal{D}_{(A_S)_{S \subset \{0, \dots, n\}}}$ be an n -Kani set over $(A_S)_{S \subset \{0, \dots, n\}}$, and let $D_{(\phi_{\{0\}}, \dots, \phi_{\{n\}})}$ be an element in $\mathcal{D}_{(A_S)_{S \subset \{0, \dots, n\}}}$. We put $D := D_{(\phi_{\{0\}}, \dots, \phi_{\{n\}})}$. We now construct endomorphisms of A_{odd} associated with D by using the matrices $M_1^{(a,b)}, \dots, M_{8a+2^b-1}^{(a,b)}$. We define homomorphisms $\psi_{i,k,l,D}: A_l \rightarrow A_k$ for $i = 1, \dots, n$ and $k, l = 1, \dots, 2^n$ by

$$\psi_{i,k,l,D} = \begin{cases} S_l * \phi_{\{0,i\}} & (\text{if } S_l \cup \{0, i\} = S_k) \\ ((S_k \setminus \{0\})_* \phi_{\{0\}}) \circ ((S_l \setminus \{i\})_* \hat{\phi}_{\{i\}}) & (\text{if } S_l \cup \{0\} = S_k \cup \{i\}) \\ ((S_k \setminus \{i\})_* \phi_{\{i\}}) \circ ((S_l \setminus \{0\})_* \hat{\phi}_{\{0\}}) & (\text{if } S_k \cup \{0\} = S_l \cup \{i\}) \\ S_k * \hat{\phi}_{\{0,i\}} & (\text{if } S_k \cup \{0, i\} = S_l) \\ 0 & (\text{otherwise}) \end{cases}.$$

We denote the endomorphism $(\psi_{i,k,l,D})_{k,l}$ by $\Psi'_{i,D}$. We also put $\Psi'_{0,D} = [d_0]_{A_{odd}}$. We construct endomorphisms $\Psi_1, \dots, \Psi_{n,D}$ of A_{odd} by $\Psi_{i,D} = \Psi'_{i,D} \odot M_i^{(a,b)}$, where $\odot$ denotes the Hadamard product. For $i = n+1, \dots, 8a+2^b-1$, we put $\Psi'_{i,D}$ by

$$\Psi'_{i,D} = \begin{cases} \Psi'_{i-n-1,D} \circ \Psi'_{i-n,D} \circ \Psi'_{i-n+1,D} & (\text{if } i \equiv b+1 \pmod{4}) \\ \Psi'_{i-n-2,D} \circ \Psi'_{i-n,D} \circ \Psi'_{i-n+1,D} & (\text{if } i \equiv b+2 \pmod{4}) \\ \Psi'_{i-n-3,D} \circ \Psi'_{i-n-2,D} \circ \Psi'_{i-n,D} & (\text{if } i \equiv b+3 \pmod{4}) \\ \Psi'_{i-n-3,D} \circ \Psi'_{i-n-2,D} \circ \Psi'_{i-n-1,D} & (\text{if } i \equiv b+4 \pmod{4}) \end{cases},$$

and define $\Psi_{i,D}$ by $\Psi_{i,D} = \Psi'_{i,D} \odot M_i^{(a,b)}$.

We summarize some properties of $\Psi_1, \dots, \Psi_{8a+2^b-1,D}$ in the following lemma.

Lemma 7 *The endomorphisms $\Psi_1, \dots, \Psi_{8a+2^b-1,D}$ satisfy the following properties:*

1. *If $\Psi_{i_1,D}, \dots, \Psi_{i_k,D}$ are all nonzero, with $1 \leq k \leq 8a+2^b-1$ and indices satisfying $1 \leq i_1 < \dots < i_k \leq 8a+2^b-1$, then $\text{Id}_{A_{odd}}, \Psi_{i_1,D}, \dots, \Psi_{i_k,D}$ are $\mathbb{Z}$ -linearly independent in the endomorphism ring of A_{odd} ,*
2. *$\Psi_{i,D} \circ \Psi_{j,D} = -\Psi_{j,D} \circ \Psi_{i,D}$ with $i, j = 1, \dots, 8a+2^b-1$ and $i \neq j$,*
3. *$\hat{\Psi}_{i,D} = -\Psi_{i,D}$ for all $i = 1, \dots, 8a+2^b-1$,*

4. $\Psi_{i,D}$ is an f'_i -isogeny, where

$$f'_i = \begin{cases} d_0 d_i & (\text{if } i = 1, \dots, n) \\ d_0^3 d_{i-n-1} d_{i-n} d_{i-n+1} & (\text{if } i > n \text{ and } i \equiv b+1 \pmod{4}) \\ d_0^3 d_{i-n-2} d_{i-n} d_{i-n+1} & (\text{if } i > n \text{ and } i \equiv b+2 \pmod{4}) \\ d_0^3 d_{i-n-3} d_{i-n-2} d_{i-n} & (\text{if } i > n \text{ and } i \equiv b+3 \pmod{4}) \\ d_0^3 d_{i-n-3} d_{i-n-2} d_{i-n-1} & (\text{if } i > n \text{ and } i \equiv b+4 \pmod{4}) \end{cases}.$$

Proof We assume that $\Psi_{i,D} \neq 0$ for some $i = 1, \dots, n$. The construction of $\Psi'_{i,D}$ and Lemma 6 imply that the (k, l) -th entry of $\Psi_{i,D}$ is zero if and only if $\epsilon_{i,k,l}^{(n)} = 0$. Moreover, from Lemma 2, this also holds if i is larger than n . This shows that $\text{Id}_{A_{\text{odd}}}, \Psi_{i_1,D}, \dots, \Psi_{i_k,D}$ are $\mathbb{Z}$ -linearly independent if $\Psi_{i_1,D}, \dots, \Psi_{i_k,D} \neq 0$, because $I_{2^n}, \overline{M}_1^{(a,b)}, \dots, \overline{M}_{8a+2^b-1}^{(a,b)}$ are $\mathbb{F}_2$ -linearly independent from Equation (1).

We prove that $\Psi'_{i,D} \circ \Psi'_{j,D} = \Psi'_{j,D} \circ \Psi'_{i,D}$ for $i, j = 1, \dots, 8a+2^b-1$. We put $\Psi'_{i,D} \circ \Psi'_{j,D} = (\psi_{(i,j),k,l,D})_{k,l}$ for $i, j = 1, \dots, n$. Here, $\psi_{(i,j),k,l,D}$ is a homomorphism from S_l to S_k . It follows from the tedious computation that

$$\begin{aligned} \psi_{(i,j),k,l,D} &= \sum_{m=1}^{2^n} \psi_{i,k,m,D} \circ \psi_{j,m,l,D} \\ &= \begin{cases} [d_0]_{A_{\text{odd}}} \circ S_l * \phi_{\{i,j\}} & (\text{if } S_l \cup \{i,j\} = S_k) \\ [d_0]_{A_{\text{odd}}} \circ ((S_k \setminus \{i\})_* \phi_{\{i\}}) \circ ((S_l \setminus \{j\})_* \hat{\phi}_{\{j\}}) & (\text{if } S_l \cup \{i\} = S_k \cup \{j\}) \\ [d_0]_{A_{\text{odd}}} \circ ((S_k \setminus \{j\})_* \phi_{\{j\}}) \circ ((S_l \setminus \{i\})_* \hat{\phi}_{\{i\}}) & (\text{if } S_k \cup \{i\} = S_l \cup \{j\}) \\ [d_0]_{A_{\text{odd}}} \circ S_k * \hat{\phi}_{\{i,j\}} & (\text{if } S_k \cup \{i,j\} = S_l) \end{cases}. \end{aligned} \quad (3)$$

Thus, it is easy to see that $\Psi'_{i,D} \circ \Psi'_{j,D} = \Psi'_{j,D} \circ \Psi'_{i,D}$ for $i, j = 1, \dots, n$. In the case of $i > n$, the homomorphism $\Psi'_{i,D}$ is defined as a product of three of $\Psi'_{0,D} = [d_0]_{A_{\text{odd}}}, \Psi'_{1,D}, \dots, \Psi'_{n,D}$. Therefore, it holds that $\Psi'_{i,D} \circ \Psi'_{j,D} = \Psi'_{j,D} \circ \Psi'_{i,D}$ for $i, j = 1, \dots, 8a+2^b-1$.

It follows from the structures of $\Psi'_{i,D}$ and $M_i^{(a,b)}$ that

$$(\Psi'_{i,D} \odot M_i^{(a,b)}) \odot (\Psi'_{j,D} \odot M_j^{(a,b)}) = (\Psi'_{i,D} \odot \Psi'_{j,D}) \odot (M_i^{(a,b)} \cdot M_j^{(a,b)})$$

for $i, j = 1, \dots, 8a+2^b-1$. As shown in the previous paragraph, two endomorphisms $\Psi'_{i,D}$ and $\Psi'_{j,D}$ are commutative; however, it holds that $M_i^{(n)} \cdot M_j^{(n)} = -M_j^{(n)} \cdot M_i^{(n)}$ with $i, j = 1, \dots, 8a+2^b-1$ and $i \neq j$. Therefore, we have $\Psi_{i,D} \circ \Psi_{j,D} = -\Psi_{j,D} \circ \Psi_{i,D}$.

From the definition of $\psi_{i,k,l,D}$, we have $\hat{\psi}_{i,k,l,D} = \psi_{i,l,k,D}$. Thus, it is easy to check that $\hat{\Psi}'_{i,D} = \Psi'_{i,D}$. Therefore, since we have ${}^t M_i^{(a,b)} = -M_i^{(a,b)}$ for $i = 1, \dots, n$, it holds that $\hat{\Psi}_{i,D} = -\Psi_{i,D}$. Since we have $\Psi'_{i,D} \circ \Psi'_{j,D} = \Psi'_{j,D} \circ \Psi'_{i,D}$, it also holds that $\hat{\Psi}_{i,D} = -\Psi_{i,D}$ for $i = n+1, \dots, 8a+2^b-1$.

Equation (3) implies that $\Psi'_{i,D} \circ \Psi'_{i,D} = [d_0 d_i]_{A_{\text{odd}}}$ for $i = 1, \dots, n$. Moreover, we have $\Psi'_{0,D} \circ \Psi'_{0,D} = [d_0^2]_{A_{\text{odd}}}$. Therefore, it holds that

$$-\Psi_{i,D}^2 = \begin{cases} [d_0 d_i]_{A_{\text{odd}}} & (\text{if } i = 1, \dots, n) \\ [d_0^3 d_{i-n-1} d_{i-n} d_{i-n+1}]_{A_{\text{odd}}} & (\text{if } i > n \text{ and } i \equiv b+1 \pmod{4}) \\ [d_0^3 d_{i-n-2} d_{i-n} d_{i-n+1}]_{A_{\text{odd}}} & (\text{if } i > n \text{ and } i \equiv b+2 \pmod{4}) \\ [d_0^3 d_{i-n-3} d_{i-n-2} d_{i-n}]_{A_{\text{odd}}} & (\text{if } i > n \text{ and } i \equiv b+3 \pmod{4}) \\ [d_0^3 d_{i-n-3} d_{i-n-2} d_{i-n-1}]_{A_{\text{odd}}} & (\text{if } i > n \text{ and } i \equiv b+4 \pmod{4}) \end{cases}.$$

Since $\hat{\Psi}_{i,D} \circ \Psi_{i,D} = \Psi_{i,D} \circ \hat{\Psi}_{i,D} = -\Psi_{i,D}^2$, this completes the proof of the assertion. $\square$

We now construct homomorphisms from A_{odd} to A_{even} by using endomorphisms $\Psi_{1,D}, \dots, \Psi_{8a+2^b-1,D}$. To obtain homomorphisms, we consider a composition of $\Psi_{i,D}$ and an appropriate homomorphism from A_{odd} to A_{even} .

We now assume that $d_0 \neq 0$ for simplicity. We can treat the case of $d_0 = 0$ in Remark 5. Let $\Phi_{0,D}: A_{\text{odd}} \rightarrow A_{\text{even}}$ be a homomorphism from A_{odd} to A_{even} represented by the matrix $(\phi_{0,k,l,D})_{k,l}$, where $\phi_{0,k,l,D}$ is a homomorphism $\phi_{0,k,l,D}: A_l \rightarrow A'_k$ defined by

$$\phi_{0,k,l,D} = \begin{cases} S_{l*} \phi_{\{0\}} & (\text{if } S'_k \ominus S_l = \{0\} \text{ and } 0 \notin S_l) \\ S'_{k*} \hat{\phi}_{\{0\}} & (\text{if } S'_k \ominus S_l = \{0\} \text{ and } 0 \in S_l) \\ 0 & (\text{otherwise}) \end{cases}.$$

We consider homomorphisms $\Phi_{0,D} \circ \Psi_{i,D}$ for $i = 1, \dots, 8a+2^b-1$. From Equation (3) and the direct calculation, we can see that $\Phi_{0,D} \circ \Psi_{i,D}$ factors through $[d_0]_{A_{\text{odd}}}$ if $i = 1, \dots, n$, and factors through $[d_0^2]_{A_{\text{odd}}}$ if $i > n$. Therefore, we can define homomorphisms $\Phi_{1,D}, \dots, \Phi_{8a+2^b-1,D}$ by

$$\Phi_{i,D} = \begin{cases} \frac{1}{d_0} \Phi_{0,D} \circ \Psi_{i,D} & (\text{if } i = 1, \dots, n) \\ \frac{1}{d_0^2} \Phi_{0,D} \circ \Psi_{i,D} & (\text{if } i = n+1, \dots, 8a+2^b-1) \end{cases}.$$

Lemma 7 derives some properties of the homomorphisms $\Phi_{0,D}, \dots, \Phi_{8a+2^b-1,D}$. We summarize these properties in the following lemma.

Lemma 8 *The homomorphisms $\Phi_{0,D}, \Phi_{1,D}, \dots, \Phi_{8a+2^b-1,D}$ satisfy the following properties:*

1. *If $\Phi_{i_1,D}, \dots, \Phi_{i_k,D}$ are all nonzero, with $1 \leq k \leq 8a+2^b$ and indices satisfying $0 \leq i_1 < \dots < i_k \leq 8a+2^b-1$, then $\Phi_{i_1,D}, \dots, \Phi_{i_k,D}$ are $\mathbb{Z}$ -linearly independent in $\text{Hom}(A_{\text{odd}}, A_{\text{even}})$,*
2. *It holds that $\Phi_{i,D} \circ \hat{\Phi}_{j,D} = -\Phi_{j,D} \circ \hat{\Phi}_{i,D}$ and $\hat{\Phi}_{i,D} \circ \Phi_{j,D} = -\hat{\Phi}_{j,D} \circ \Phi_{i,D}$ with $i, j = 1, \dots, 8a+2^b-1$ and $i \neq j$,*
3. *It holds that $\Phi_{0,D} \circ \hat{\Phi}_{i,D} = -\Phi_{i,D} \circ \hat{\Phi}_{0,D}$ and $\hat{\Phi}_{0,D} \circ \Phi_{i,D} = -\hat{\Phi}_{i,D} \circ \Phi_{0,D}$ for all $i = 1, \dots, 8a+2^b-1$,*

4. $\Phi_{i,D}$ is an f_i -isogeny, where

$$f_i = \begin{cases} d_i & (\text{if } i = 0, \dots, n) \\ d_{i-n-1}d_{i-n}d_{i-n+1} & (\text{if } i > n \text{ and } i \equiv b+1 \pmod{4}) \\ d_{i-n-2}d_{i-n}d_{i-n+1} & (\text{if } i > n \text{ and } i \equiv b+2 \pmod{4}) \\ d_{i-n-3}d_{i-n-2}d_{i-n} & (\text{if } i > n \text{ and } i \equiv b+3 \pmod{4}) \\ d_{i-n-3}d_{i-n-2}d_{i-n-1} & (\text{if } i > n \text{ and } i \equiv b+4 \pmod{4}) \end{cases}.$$

Proof The assertion follows from Lemma 7 and two properties of $\Phi_{0,D}$. The first property is that $\Phi_{0,D}$ is a d_0 -isogeny, and the second property is that $\Phi_{0,D}$ is represented by a diagonal matrix with all nonzero diagonal entries. $\square$

Remark 5 We now focus on the case $\phi_{\{0\}} = 0$. It is easy to see that, if $i \notin \{0, n+1, n+2, n+3\}$, then the homomorphism $\Phi_{i,D}$ does not depend on the isogeny $\phi_{\{0\}}$. Therefore, even if $\phi_{\{0\}} = 0$, we can define $\Phi_{i,D}$ for $i \notin \{0, n+1, n+2, n+3\}$ by taking an appropriate isogeny $\phi'_{\{0\}}: A_\emptyset \rightarrow A_{\{0\}}$ instead of $\phi_{\{0\}}$. In the case of $i \in \{0, n+1, n+2, n+3\}$, we can define $\Phi_{i,D}$ by $\Phi_{i,D} = 0$.

We now obtain the main theorem of our result. This theorem provides an explicit construction of an n -Kani relation. In other words, this theorem is a generalization of Kani's lemma.

Theorem 9 Let n be a positive integer, and let a, b be integers such that $n = 4a + b$ and $b = 0, 1, 2, 3$. Let f be a polynomial in $\mathbb{Z}[x_0, \dots, x_n]$ defined by

$$\begin{aligned} f(x_0, \dots, x_n) = & \sum_{i=0}^n \alpha_i^2 x_i + \sum_{\substack{i \in \{n+1, \dots, 8a+2^b-1\} \\ i \equiv b+1 \pmod{4}}} \alpha_i^2 x_{i-n-1} x_{i-n} x_{i-n+1} \\ & + \sum_{\substack{i \in \{n+1, \dots, 8a+2^b-1\} \\ i \equiv b+2 \pmod{4}}} \alpha_i^2 x_{i-n-2} x_{i-n} x_{i-n+1} \\ & + \sum_{\substack{i \in \{n+1, \dots, 8a+2^b-1\} \\ i \equiv b+3 \pmod{4}}} \alpha_i^2 x_{i-n-3} x_{i-n-2} x_{i-n} \\ & + \sum_{\substack{i \in \{n+1, \dots, 8a+2^b-1\} \\ i \equiv b+4 \pmod{4}}} \alpha_i^2 x_{i-n-3} x_{i-n-2} x_{i-n-1} \end{aligned}$$

with $\alpha_0, \dots, \alpha_{8a+2^b-1} \in \mathbb{Z}$ and $\alpha_0, \dots, \alpha_n \neq 0$. Then, there is an n -Kani relation over $(A_S)_{S \subset \{0, \dots, n\}}$ for $(A_{\text{odd}}, A_{\text{even}})$ associated with f given by

$$\begin{aligned} R_{(A_S)_{S \subset \{0, \dots, n\}}} : \mathcal{D}_{(A_S)_{S \subset \{0, \dots, n\}}} & \longrightarrow \text{Hom}(A_{\text{odd}}, A_{\text{even}}) / \sim_d \\ D & \longmapsto \sum_{i=0}^{8a+2^b-1} \alpha_i \cdot \Phi_{i,D}. \end{aligned}$$

Proof We first show that $R_{(A_S)_{S \subset \{0, \dots, n\}}}$ is well-defined. Let D' be an n -Kani diagram over $(A'_S)_{S \subset \{0, \dots, n\}}$ equivalent to $D \in \mathcal{D}_{(A_S)_{S \subset \{0, \dots, n\}}}$, where A'_S is a principally polarized abelian variety with $A_S \cong A'_S$. We put $D = D_{(\phi_{\{0\}}, \dots, \phi_{\{n\}})}$ and $D' = D_{(\phi'_{\{0\}}, \dots, \phi'_{\{n\}})}$. For $S \subset \{0, \dots, n\}$, let $\lambda_S: A_S \rightarrow A'_S$ be isomorphisms satisfying $S_* \phi'_{\{i\}} \circ \lambda_S = \lambda_{S \cup \{i\}} \circ S_* \phi_{\{i\}}$ for all $i = 0, \dots, n$. We have $S_* \hat{\phi}'_{\{i\}} \circ \lambda_{S \cup \{i\}} = \lambda_S \circ S_* \hat{\phi}_{\{i\}}$. Therefore, the constructions of $\Phi_{i,D}$ and $\Phi_{i,D'}$ derive that

$$\Phi_{i,D'} \circ \text{diag}((\lambda_{S_k})_k) = \text{diag}((\lambda_{S'_k})_k) \circ \Phi_{i,D}$$

for $i = 0, \dots, 8a + 2^b - 1$. Thus, it holds that

$$\alpha_0 \cdot \Phi_{0,D} + \dots + \alpha_{8a+2^b-1} \cdot \Phi_{8a+2^b-1,D} \sim_d \alpha_0 \cdot \Phi_{0,D'} + \dots + \alpha_{8a+2^b-1} \cdot \Phi_{8a+2^b-1,D'}.$$

This proves the well-definedness of $R_{(A_S)_{S \subset \{0, \dots, n\}}}$.

We finally show that the map $R_{(A_S)_{S \subset \{0, \dots, n\}}}$ satisfies the desired conditions of n -Kani relations. That is, we prove the following properties (A-C):

- A. The (k, l) -th entry $r_{k,l}$ of $R_{(A_S)_{S \subset \{0, \dots, n\}}}$ is represented by a map such that, for $\beta_0, \dots, \beta_n \in \mathbb{Z}$,

$$r_{k,l}(D_{(\beta_0 \phi_{\{0\}}, \dots, \beta_n \phi_{\{n\}})}) = \left(\prod_{i \in S \ominus S'} \beta_i \right) \cdot r_{k,l}(D_{(\phi_{\{0\}}, \dots, \phi_{\{n\}})}),$$

- B. $R_{(A_S)_{S \subset \{0, \dots, n\}}}$ is injective,
 C. $\sum_{i=0}^{8a+2^b-1} \alpha_i \cdot \Phi_{i,D}$ is an $f(d_0, \dots, d_n)$ -isogeny.

We first show the property (A). From the constructions $\Phi_{0,D}, \dots, \Phi_{8a+2^b-1,D}$, the (k, l) -th entry of $\Phi_{i,D}$ is represented by

$$\epsilon \cdot ((S'_k \setminus T')_* \phi_{T'} \circ (S_l \setminus T)_* \hat{\phi}_T),$$

where $\epsilon = 0, 1, -1$, and T, T' are sets with $T \sqcup T' = S_l \ominus S'_k$, $T \subset S_l$, and $T' \subset S'_k$. Therefore, the property (A) holds.

We next prove the property (B), the injectivity of $R_{(A_S)_{S \subset \{0, \dots, n\}}}$. For an n -Kani diagram $D = D_{(\phi_{\{0\}}, \dots, \phi_{\{n\}})}$ over $(A_S)_{S \subset \{0, \dots, n\}}$, we assume that $\phi_{\{i\}} \neq 0$ for some $i = 0, \dots, n$. Since $\alpha_i \neq 0$, we have $\alpha_i \cdot \Phi_{i,D} \neq 0$. The property (1) in Lemma 7 implies that $R_{(A_S)_{S \subset \{0, \dots, n\}}}(D_{(\phi_{\{0\}}, \dots, \phi_{\{n\}})}) \neq 0$. Therefore, the map $R_{(A_S)_{S \subset \{0, \dots, n\}}}$ is injective.

The property (C) can be shown by the properties (2), (3), and (4) of $\Phi_0, \dots, \Phi_{8a+2^b-1}$ in Lemma 7. We put $\Phi_D := \sum_{i=0}^{8a+2^b-1} \alpha_i \cdot \Phi_{i,D}$. It follows from the properties (2) and (3) that

$$\hat{\Phi}_D \circ \Phi_D = \left(\sum_{i=0}^{8a+2^b-1} \alpha_i \cdot \hat{\Phi}_{i,D} \right) \circ \left(\sum_{i=0}^{8a+2^b-1} \alpha_i \cdot \Phi_{i,D} \right) = \sum_{i=0}^{8a+2^b-1} \alpha_i^2 \cdot [f_i]_{A_{\text{odd}}}.$$

Therefore, from the property (4), we have $\hat{\Phi}_D \circ \Phi_D = [f(d_0, \dots, d_n)]_{A_{\text{odd}}}$. Similarly, we also have $\Phi_D \circ \hat{\Phi}_D = [f(d_0, \dots, d_n)]_{A_{\text{even}}}$. Hence, Φ_D is an $f(d_0, \dots, d_n)$ -isogeny. $\square$

Remark 6 One may consider a polynomial f without satisfying the condition $\alpha_0, \dots, \alpha_n \neq 0$. In this case, the map R in Theorem 9 is not injective. Indeed, if $\alpha_i = 0$ for some $i = 0, \dots, n$, then we can check that $R(D_{(0, \dots, 0, \phi_{\{i\}}, 0, \dots, 0)}) = 0$ for any isogeny $\phi_{\{i\}}: A_\emptyset \rightarrow A_{\{i\}}$.

4.3 Examples of n -Kani relations

In this subsection, we demonstrate our result for $n = 1, 2, 3$. That is, we explicitly write down 1, 2, and 3-Kani relations based on Theorem 9.

If $n = 1$, Theorem 9 gives the same isogeny derived from the original Kani's lemma. If $n = 2, 3$, Theorem 9 appears as a nontrivial generalization of Kani's lemma.

4.3.1 1-Kani relation.

Let d_0, d_1 be positive and pairwise coprime integers. Let $(A_S)_{S \subset \{0,1\}}$ be principally polarized abelian varieties, and let $D_{(\phi_{\{0\}}, \phi_{\{1\}})}$ be a 1-Kani diagram over $(A_S)_{S \subset \{0,1\}}$ of order (d_0, d_1) . Then, there is a 1-Kani isogeny

$$\Phi: A_{\{0\}} \times A_{\{1\}} \longrightarrow A_{\emptyset} \times A_{\{0,1\}}$$

of order $N = \alpha_0^2 d_0 + \alpha_1^2 d_1$ given by

$$\Phi = \begin{pmatrix} \alpha_0 \hat{\phi}_{\{0\}} & \alpha_1 \hat{\phi}_{\{1\}} \\ -\alpha_1 \{0\} * \phi_{\{1\}} & \alpha_0 \{1\} * \phi_{\{0\}} \end{pmatrix},$$

where $\alpha_0, \alpha_1 \in \mathbb{Z}$ and $\alpha_0, \alpha_1 \neq 0$.

The kernel of the isogeny Φ is given by $\hat{\Phi}((A_{\{0\}} \times A_{\{1\}})[N])$. In particular, if N is coprime to d_0 and d_1 , then

$$\ker \Phi = \left\langle \begin{pmatrix} \alpha_0 \phi_{\{0\}}(P) \\ \alpha_1 \phi_{\{1\}}(P) \end{pmatrix} \mid P \in A_{\emptyset}[N] \right\rangle.$$

4.3.2 2-Kani relation.

Let d_0, d_1, d_2 be positive and pairwise coprime integers. Let $(A_S)_{S \subset \{0,1,2\}}$ be principally polarized abelian varieties, and let $D_{(\phi_{\{0\}}, \phi_{\{1\}}, \phi_{\{2\}})}$ be a 2-Kani diagram over $(A_S)_{S \subset \{0,1,2\}}$ of order (d_0, d_1, d_2) . Then, there is a 2-Kani isogeny

$$\Phi: A_{\{0\}} \times A_{\{1\}} \times A_{\{2\}} \times A_{\{0,1,2\}} \longrightarrow A_{\emptyset} \times A_{\{0,1\}} \times A_{\{0,2\}} \times A_{\{1,2\}}$$

of order $N = \alpha_0^2 d_0 + \alpha_1^2 d_1 + \alpha_2^2 d_2 + \alpha_3^2 d_0 d_1 d_2$ represented by the matrix

$$\Phi = \begin{pmatrix} \alpha_0 \hat{\phi}_{\{0\}} & \alpha_1 \hat{\phi}_{\{1\}} & \alpha_2 \hat{\phi}_{\{2\}} & \alpha_3 \hat{\phi}_{\{0,1,2\}} \\ -\alpha_1 \{0\} * \phi_{\{1\}} & \alpha_0 \{1\} * \phi_{\{0\}} & -\alpha_3 \phi_{\{0,1\}} \circ \hat{\phi}_{\{2\}} & \alpha_2 \{0,1\} * \hat{\phi}_{\{2\}} \\ -\alpha_2 \{0\} * \phi_{\{2\}} & \alpha_3 \phi_{\{0,2\}} \circ \hat{\phi}_{\{1\}} & \alpha_0 \{2\} * \phi_{\{0\}} & -\alpha_1 \{0,2\} * \hat{\phi}_{\{1\}} \\ -\alpha_3 \phi_{\{1,2\}} \circ \hat{\phi}_{\{0\}} & -\alpha_2 \{1\} * \phi_{\{2\}} & \alpha_1 \{2\} * \phi_{\{1\}} & \alpha_0 \{1,2\} * \hat{\phi}_{\{0\}} \end{pmatrix},$$

where $\alpha_0, \alpha_1, \alpha_2, \alpha_3 \in \mathbb{Z}$ and $\alpha_0, \alpha_1, \alpha_2 \neq 0$.

The kernel of the isogeny is

$$\ker \Phi = \hat{\Phi}((A_{\{0\}} \times A_{\{1\}} \times A_{\{2\}} \times A_{\{0,1,2\}})[N]).$$

In particular, if N is coprime to d_0, d_1 , and d_2 , then $\ker \Phi$ is given by

$$\ker \Phi = \left\langle \begin{pmatrix} \alpha_0 \phi_{\{0\}}(P) \\ \alpha_1 \phi_{\{1\}}(P) \\ \alpha_2 \phi_{\{2\}}(P) \\ \alpha_3 \phi_{\{0,1,2\}}(P) \end{pmatrix}, \begin{pmatrix} -\alpha_1 d_1 \phi_{\{0\}}(P) \\ \alpha_0 d_0 \phi_{\{1\}}(P) \\ -\alpha_3 d_0 d_1 \phi_{\{2\}}(P) \\ \alpha_2 \phi_{\{0,1,2\}}(P) \end{pmatrix} \middle| P \in A_\emptyset[N] \right\rangle.$$

4.3.3 3-Kani relation.

Let d_0, d_1, d_2, d_3 be positive and pairwise coprime integers. Let $(A_S)_{S \subset \{0,1,2,3\}}$ be principally polarized abelian varieties, and let $D_{(\phi_{\{0\}}, \phi_{\{1\}}, \phi_{\{2\}}, \phi_{\{3\}})}$ be a 3-Kani diagram over $(A_S)_{S \subset \{0,1,2,3\}}$ of order (d_0, d_1, d_2, d_3) . Then, there is a 3-Kani isogeny

$$\begin{aligned} \Phi: A_{\{0\}} \times A_{\{1\}} \times A_{\{2\}} \times A_{\{0,1,2\}} \times A_{\{3\}} \times A_{\{0,1,3\}} \times A_{\{0,2,3\}} \times A_{\{1,2,3\}} \\ \longrightarrow A_\emptyset \times A_{\{0,1\}} \times A_{\{0,2\}} \times A_{\{1,2\}} \times A_{\{0,3\}} \times A_{\{1,3\}} \times A_{\{2,3\}} \times A_{\{0,1,2,3\}} \end{aligned}$$

of order

$$N = \alpha_0^2 d_0 + \alpha_1^2 d_1 + \alpha_2^2 d_2 + \alpha_3^2 d_3 + \alpha_4^2 d_0 d_1 d_2 + \alpha_5^2 d_0 d_2 d_3 + \alpha_6^2 d_0 d_1 d_3 + \alpha_7^2 d_1 d_2 d_3$$

represented by the 8×8 -matrix

$$\begin{pmatrix} \alpha_0 \hat{\phi}_{\{0\}} & \alpha_1 \hat{\phi}_{\{1\}} & \alpha_2 \hat{\phi}_{\{2\}} & \alpha_4 \hat{\phi}_{\{0,1,2\}} \\ -\alpha_1 \{0\} * \hat{\phi}_{\{1\}} & \alpha_0 \{1\} * \hat{\phi}_{\{0\}} & -\alpha_4 \phi_{\{0,1\}} \circ \hat{\phi}_{\{2\}} & \alpha_2 \{0,1\} * \hat{\phi}_{\{2\}} \\ -\alpha_2 \{0\} * \hat{\phi}_{\{2\}} & \alpha_4 \phi_{\{0,2\}} \circ \hat{\phi}_{\{1\}} & \alpha_0 \{2\} * \hat{\phi}_{\{0\}} & -\alpha_1 \{0,2\} * \hat{\phi}_{\{1\}} \\ -\alpha_4 \phi_{\{1,2\}} \circ \hat{\phi}_{\{0\}} & -\alpha_2 \{1\} * \hat{\phi}_{\{2\}} & \alpha_1 \{2\} * \hat{\phi}_{\{1\}} & \alpha_0 \{1,2\} * \hat{\phi}_{\{0\}} \\ -\alpha_3 \{0\} * \hat{\phi}_{\{3\}} & \alpha_6 \phi_{\{0,3\}} \circ \hat{\phi}_{\{1\}} & \alpha_5 \phi_{\{0,3\}} \circ \hat{\phi}_{\{2\}} & \alpha_7 \{0\} * (\phi_{\{3\}} \circ \hat{\phi}_{\{1,2\}}) \\ -\alpha_6 \phi_{\{1,3\}} \circ \hat{\phi}_{\{0\}} & -\alpha_3 \{1\} * \hat{\phi}_{\{3\}} & \alpha_7 \phi_{\{1,3\}} \circ \hat{\phi}_{\{2\}} & -\alpha_5 \{1\} * (\phi_{\{3\}} \circ \hat{\phi}_{\{0,2\}}) \\ -\alpha_5 \phi_{\{2,3\}} \circ \hat{\phi}_{\{0\}} & -\alpha_7 \phi_{\{2,3\}} \circ \hat{\phi}_{\{1\}} & -\alpha_3 \{2\} * \hat{\phi}_{\{3\}} & \alpha_6 \{2\} * (\phi_{\{3\}} \circ \hat{\phi}_{\{0,1\}}) \\ -\alpha_7 \{0\} * \hat{\phi}_{\{1,2,3\}} & \alpha_5 \{1\} * \hat{\phi}_{\{0,2,3\}} & -\alpha_6 \{2\} * \hat{\phi}_{\{0,1,3\}} & -\alpha_3 \{0,1,2\} * \hat{\phi}_{\{3\}} \end{pmatrix} \begin{pmatrix} \alpha_3 \hat{\phi}_{\{3\}} & \alpha_6 \hat{\phi}_{\{0,1,3\}} & \alpha_5 \hat{\phi}_{\{0,2,3\}} & \alpha_7 \hat{\phi}_{\{1,2,3\}} \\ -\alpha_6 \phi_{\{0,1\}} \circ \hat{\phi}_{\{3\}} & \alpha_3 \{0,1\} * \hat{\phi}_{\{3\}} & \alpha_7 \{0\} * (\phi_{\{1\}} \circ \hat{\phi}_{\{2,3\}}) & -\alpha_5 \{1\} * (\phi_{\{0\}} \circ \hat{\phi}_{\{2,3\}}) \\ -\alpha_5 \phi_{\{0,2\}} \circ \hat{\phi}_{\{3\}} & -\alpha_7 \{0\} * (\phi_{\{2\}} \circ \hat{\phi}_{\{1,3\}}) & \alpha_3 \{0,2\} * \hat{\phi}_{\{3\}} & \alpha_6 \{2\} * (\phi_{\{0\}} \circ \hat{\phi}_{\{1,3\}}) \\ -\alpha_7 \phi_{\{1,2\}} \circ \hat{\phi}_{\{3\}} & \alpha_5 \{1\} * (\phi_{\{2\}} \circ \hat{\phi}_{\{0,3\}}) & -\alpha_6 \{2\} * (\phi_{\{1\}} \circ \hat{\phi}_{\{0,3\}}) & \alpha_3 \{1,2\} * \hat{\phi}_{\{3\}} \\ \alpha_0 \{3\} * \hat{\phi}_{\{0\}} & -\alpha_1 \{0,3\} * \hat{\phi}_{\{1\}} & -\alpha_2 \{0,3\} * \hat{\phi}_{\{2\}} & -\alpha_4 \{3\} * (\phi_{\{0\}} \circ \hat{\phi}_{\{1,2\}}) \\ \alpha_1 \{3\} * \hat{\phi}_{\{1\}} & \alpha_0 \{1,3\} * \hat{\phi}_{\{0\}} & \alpha_4 \{3\} * (\phi_{\{1\}} \circ \hat{\phi}_{\{0,2\}}) & -\alpha_2 \{1,3\} * \hat{\phi}_{\{2\}} \\ \alpha_2 \{3\} * \hat{\phi}_{\{2\}} & -\alpha_4 \{3\} * (\phi_{\{2\}} \circ \hat{\phi}_{\{0,1\}}) & \alpha_0 \{2,3\} * \hat{\phi}_{\{0\}} & \alpha_1 \{2,3\} * \hat{\phi}_{\{1\}} \\ \alpha_4 \{3\} * \hat{\phi}_{\{0,1,2\}} & \alpha_2 \{0,1,3\} * \hat{\phi}_{\{2\}} & -\alpha_1 \{0,2,3\} * \hat{\phi}_{\{1\}} & \alpha_0 \{1,2,3\} * \hat{\phi}_{\{0\}} \end{pmatrix},$$

where $\alpha_0, \dots, \alpha_7 \in \mathbb{Z}$, and $\alpha_0, \dots, \alpha_3 \neq 0$.

The kernel of the isogeny $\ker \Phi$ is

$$\hat{\Phi}((A_{\{0\}} \times A_{\{1\}} \times A_{\{2\}} \times A_{\{0,1,2\}} \times A_{\{3\}} \times A_{\{0,1,3\}} \times A_{\{0,2,3\}} \times A_{\{1,2,3\}})[N]).$$

In particular, if N is coprime to d_0, d_1, d_2 , and d_3 , then $\ker \Phi$ is generated by

$$\begin{pmatrix} \alpha_0 \phi_{\{0\}}(P) \\ \alpha_1 \phi_{\{1\}}(P) \\ \alpha_2 \phi_{\{2\}}(P) \\ \alpha_4 \phi_{\{0,1,2\}}(P) \\ \alpha_3 \phi_{\{3\}}(P) \\ \alpha_6 \phi_{\{0,1,3\}}(P) \\ \alpha_5 \phi_{\{0,2,3\}}(P) \\ \alpha_7 \phi_{\{1,2,3\}}(P) \end{pmatrix}, \begin{pmatrix} -\alpha_1 d_1 \phi_{\{0\}}(P) \\ \alpha_0 d_0 \phi_{\{1\}}(P) \\ -\alpha_4 d_0 d_1 \phi_{\{2\}}(P) \\ \alpha_2 \phi_{\{0,1,2\}}(P) \\ -\alpha_6 d_0 d_1 \phi_{\{3\}}(P) \\ \alpha_3 \phi_{\{0,1,3\}}(P) \\ \alpha_7 d_1 \phi_{\{0,2,3\}}(P) \\ -\alpha_5 d_0 \phi_{\{1,2,3\}}(P) \end{pmatrix}, \begin{pmatrix} -\alpha_2 d_2 \phi_{\{0\}}(P) \\ \alpha_4 d_0 d_2 \phi_{\{1\}}(P) \\ \alpha_0 d_0 \phi_{\{2\}}(P) \\ -\alpha_1 \phi_{\{0,1,2\}}(P) \\ -\alpha_5 d_0 d_2 \phi_{\{3\}}(P) \\ -\alpha_7 d_2 \phi_{\{0,1,3\}}(P) \\ \alpha_3 \phi_{\{0,2,3\}}(P) \\ \alpha_6 d_0 \phi_{\{1,2,3\}}(P) \end{pmatrix}, \text{ and } \begin{pmatrix} -\alpha_4 d_1 d_2 \phi_{\{0\}}(P) \\ -\alpha_2 d_2 \phi_{\{1\}}(P) \\ \alpha_1 d_1 \phi_{\{2\}}(P) \\ \alpha_0 \phi_{\{0,1,2\}}(P) \\ -\alpha_7 d_1 d_2 \phi_{\{3\}}(P) \\ \alpha_5 d_2 \phi_{\{0,1,3\}}(P) \\ -\alpha_6 d_1 \phi_{\{0,2,3\}}(P) \\ \alpha_3 \phi_{\{1,2,3\}}(P) \end{pmatrix},$$

where $P \in A_\emptyset[N]$.

4.4 Limitation to extend Kani's lemma

The complexity in computing an N -isogeny of principally polarized abelian varieties of dimension g is known as $O((cN)^g)$ for $c \in \mathbb{Z}_{\geq 2}$ [14, Theorem 1.3]. Therefore, it is an interesting research question to determine the lower bound of dimensions of domain/codomain abelian varieties of Kani isogenies obtained from an n -Kani relation.

In this subsection, we give the answer to the above question (Theorem 10). Unfortunately, if there is an n -Kani relation to $\mathcal{I}_{(A_{dom}, A_{codom})}$, then the dimension of A_{dom} and A_{codom} is 2^n . Therefore, in our formulation, we cannot shrink the dimensions of abelian varieties appearing in some isogeny-based cryptographical contexts. For example, a four-dimensional isogeny is used in SQISignHD [6], but we cannot use three- or two-dimensional isogenies for the protocol without changing the parameters.

Theorem 10 *Let $\mathcal{S}_{dom}$ and $\mathcal{S}_{codom}$ be sets of subsets of $\{0, \dots, n\}$ with $\#\mathcal{S}_{dom} = \#\mathcal{S}_{codom}$ and $\mathcal{S}_{dom} \cap \mathcal{S}_{codom} = \emptyset$. We define A_{dom} and A_{codom} by*

$$A_{dom} := \prod_{S \in \mathcal{S}_{dom}} A_S, \quad A_{codom} := \prod_{S \in \mathcal{S}_{codom}} A_S.$$

Assume that there is an n -Kani relation

$$R : \mathcal{D}_{(A_S)_{S \subset \{0, \dots, n\}}} \longrightarrow \mathcal{I}_{(A_{dom}, A_{codom})}.$$

Then, one of the followings holds:

- $\mathcal{S}_{dom} = \mathcal{S}_{odd}^{(n)}$ and $\mathcal{S}_{codom} = \mathcal{S}_{even}^{(n)}$,
- $\mathcal{S}_{dom} = \mathcal{S}_{even}^{(n)}$ and $\mathcal{S}_{codom} = \mathcal{S}_{odd}^{(n)}$.

Proof We focus on the diagrams $D_i := D_{(0, \dots, 0, \phi_{\{i\}}, 0, \dots, 0)} \in \mathcal{D}_{(A_S)_{S \subset \{0, \dots, n\}}}$ for $i = 0, \dots, n$.

We denote the matrix that represents $R(D_i)$ by $(r_{S, S'})_{S, S'}$, where $S \in \mathcal{S}_{dom}$ and $S' \in \mathcal{S}_{codom}$.

It follows from the definition of the image matrices of R that if $r_{S, S'} \neq 0$, then $S \ominus S' = \{i\}$. Moreover, since R is injective, we have $R(D_i)^* \cdot R(D_i) \neq 0$. This implies that all rows of $R(D_i)$ have at least one non-zero element. Therefore, for any $S \in \mathcal{S}_{dom}$, we have $S \ominus \{i\}$ belongs to $\mathcal{S}_{codom}$. Similarly, for any $S' \in \mathcal{S}_{codom}$, the subset $S' \ominus \{i\}$ belongs to $\mathcal{S}_{dom}$.

From the above, it holds that, if $S \in \mathcal{S}_{dom}$, then $S \ominus \{i, j\} \in \mathcal{S}_{dom}$ for any $0 \leq i < j \leq n$. Let $T \in \mathcal{S}_{dom}$, and let T' be a subset of $\{0, \dots, n\}$. We put $T = \{i_1, \dots, i_k\}$ and $T' = \{i'_1, \dots, i'_l\}$. If $\#T \equiv \#T' \pmod{2}$, then $T' \in \mathcal{S}_{dom}$ because we have

$$T' = T \ominus \{i_1, i'_1\} \ominus \dots \ominus \{i_k, i'_k\} \ominus \{i'_{k+1}, i'_{k+2}\} \ominus \dots \ominus \{i'_{l-1}, i'_l\}.$$

Here, we assume that $k \leq l$. We can also prove the claim in the case of $k > l$ similarly. Therefore, it holds that $\mathcal{S}_{odd}^{(n)} \subset \mathcal{S}_{dom}$ or $\mathcal{S}_{even}^{(n)} \subset \mathcal{S}_{dom}$. Note that $\#\mathcal{S}_{dom}$ is less than or equal to $\frac{1}{2} \cdot \#\mathcal{S}^{(n)}$, which equals to $\#\mathcal{S}_{odd}^{(n)}$ and $\#\mathcal{S}_{even}^{(n)}$. Thus, we have $\mathcal{S}_{dom} = \mathcal{S}_{odd}^{(n)}$ or $\mathcal{S}_{dom} = \mathcal{S}_{even}^{(n)}$. The remaining of the proof derives from $\mathcal{S}_{odd}^{(n)} \sqcup \mathcal{S}_{even}^{(n)} = \mathcal{S}^{(n)}$. $\square$

5 Conclusion

We first formalized a generalization of Kani's lemma. We defined an n -Kani relation, which gives $2^n g$ -dimensional abelian varieties from a commutative diagram with 2^{n+1} abelian varieties of dimension g .

We next proved the existence of n -Kani relations, which is a generalization of Kani's lemma. To construct an n -Kani relation, we used a representation of the Clifford algebra $\mathbb{R}_{0,8a+2^b-1}$ with $n = 4a + b$ and $b = 0, 1, 2, 3$.

We also gave examples of the constructions in the cases of $n = 1, 2, 3$. Our result provides the same construction of the original Kani's lemma in $n = 1$, while it provides nontrivial extensions of Kani's lemma in $n = 2$ and 3 .

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